



Fig. 1: Thermal relief

ture. An uncontrolled iron will simply cool down. This also means that you would need different wattage irons to do a PCB if you are using unregulated irons. With a temperature-controlled soldering iron, you can easily adjust the temperature corresponding to the requirement of different joints.

So the conclusion is: Never go for cheap unregulated soldering irons. These will prove to be more expensive than temperature-controlled irons one day and you will regret it.

Now let's find out how to select the right temperature-controlled soldering iron. To decide, carefully check the specifications mentioned below and consider the recommendations.

Specification recommendations

Analogue or digital? A soldering station consists of a soldering iron whose temperature is controlled through an electrical power supply. The control may be as simple as a variable potentiometer, or little more advanced like a microprocessor-based control. The potentiometer type roughly sets the temperature of the soldering station. The microcontroller-based type controls the temperature at the heating element using the reading from a thermocouple and heats the tip according to the user-defined settings.

Digital vs analogue refers to differ-

ent ways of telling you about the tip temperature. Digital looks more professional and can actually read in finer increments. However, such precision is not really required and it has nothing to do with the type of iron. Therefore choose the digital one only when you really need it, as it is expensive.

Wattage. The wattage that you would require completely depends on the project in hand. Usually, bigger components absorb and dissipate more heat. If this rate is more than the power of the soldering iron, it starts cooling down. So the general rule is that you need more wattage for larger parts.

Additional tips

- Look for good, comfortable handle grip in the soldering iron.
- Choose a soldering iron with replaceable tips and heating element. These should be readily available too.
- A separate stand is highly useful and comfortable while working.
- A sponge and wire cleaning section are must.
- Cord length should be sufficient for your desk.
- Always go with a known brand. Cheap Chinese soldering irons don't last long.

Typically, power rating of 15-25 watts is sufficient for most electronics work. A higher wattage does not mean that the iron runs hotter—it simply means that there is more power in reserve when required. If you are just soldering small components and ICs, 15 watts will be fine, but you may have to wait a bit in between joints for the tip to recover its temperature. If you are soldering larger components, especially ones with heat sinks (like voltage regulators), you will probably need a 25W or 30W iron. For soldering larger things like 10-gauge copper wire, motor casings or large heat sinks, you need a soldering iron with 50W or higher rating.

Take a higher-wattage iron considering future requirements too. For large components and joints, you would need

Lead vs lead-free solder

The European Union now requires all commercially available electronics products to use lead-free components, including solder.

Lead is used to reduce the melting point of the alloy and with lead-free coming in, different metals need to be used that are not equally effective.

Lead-free solders are a bit harder to use as these don't flow as nicely or at as low temperature as their lead-based equivalent.

When a lead-based solder is used, good joints can immediately be seen as they are shiny and shaped like a tent. And with very little practice, you can master good soldering.

However, in case of lead-free solder, good joints are not shiny, and it is very difficult to visually detect a bad joint.

If you are not planning to sell the product in the EU, stick to a lead-based solder.

Lead-free solder alloys

Sn96/Ag4
Sn96.5/Ag3.5
Sn93.6/Ag4.7/Cu1.7
Sn95.2/Ag4/Cu0.8
Sn95.2/Ag3.9/Cu0.9
Sn95.2/Ag3.8/Cu1
Sn95.5/Ag3.5/Cu1
Sn96.2/Ag3/Cu0.7
Sn96.5/Ag3/Cu0.5
Sn96.2/Ag2.5/Cu0.8/Sb0.5
Sn99.3/Cu0.7